

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
9	(a)			First response starts at zero / booster {doesn't / already has antibodies / memory cells present} (1) antibody levels increase faster after the booster / slower after first vaccine (1) first response peak is low(er) / peak after the booster high(er) OWTTE (1) First response – antibody levels fall below threshold / antibody levels don't drop below threshold after booster (1) Antibody levels could be referred to as immune response		4		4		
	(b)			Antigens (1) stimulate antibody production by (white blood cells / lymphocytes) (1) Memory cells produced / produce antibodies very quickly if the antigen is encountered again (1)	3					
				Question 9 total	3	4		7		